# Lessons From the Autoregressive/Non-autoregressive Battle in Speech Synthesis

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# About Me

- Xu Tan (谭旭)
  - Principal Researcher and Research Manager @ Microsoft Research Asia
- Research interests
  - Speech: FastSpeech 1/2, NaturalSpeech 1/2, UniAudio (https://speechresearch.github.io/)
  - Music: Muzic project (https://github.com/microsoft/muzic)
  - Avatar: GAIA project (https://microsoft.github.io/GAIA/)
  - NLP (machine translation, LLMs)
- Homepage
  - https://www.microsoft.com/en-us/research/people/xuta/
  - https://scholar.google.com/citations?user=tob-U1oAAAAJ

# Text-to-Speech Synthesis

- Text-to-speech (TTS): generate intelligible and natural speech from text

*"speech synthesis is awesome"*

Text → TTS System → Speech

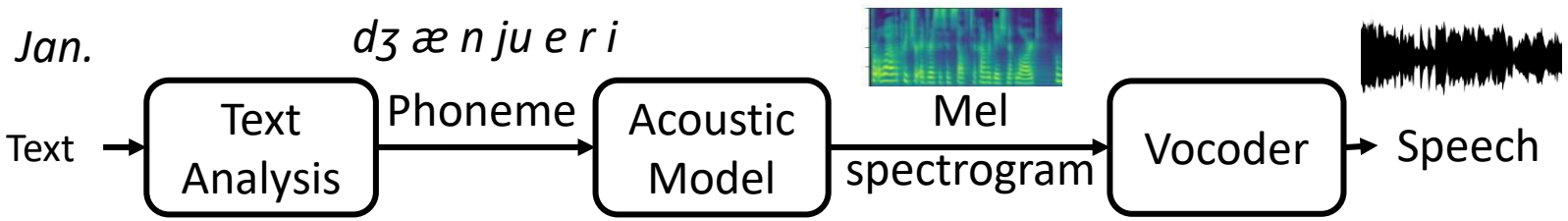
- Enabling machine to speak is an important part of AI
  - **TTS (speaking)** is as important as **ASR (listening), NLU (reading), NLG (writing)**
  - Human beings tried to build TTS systems dating back to the **12th century**

**Neural TTS** → **WaveNet (Google DeepMind)**

| 1950s | 1970s | 1990s | | 2010s | 2016 |
|---|---|---|---|---|---|
| Articulatory Synthesis | Formant Synthesis | Concatenative Synthesis | Statistical Parametric Synthesis | Neural Speech Synthesis | (Deep) Neural Speech Synthesis |

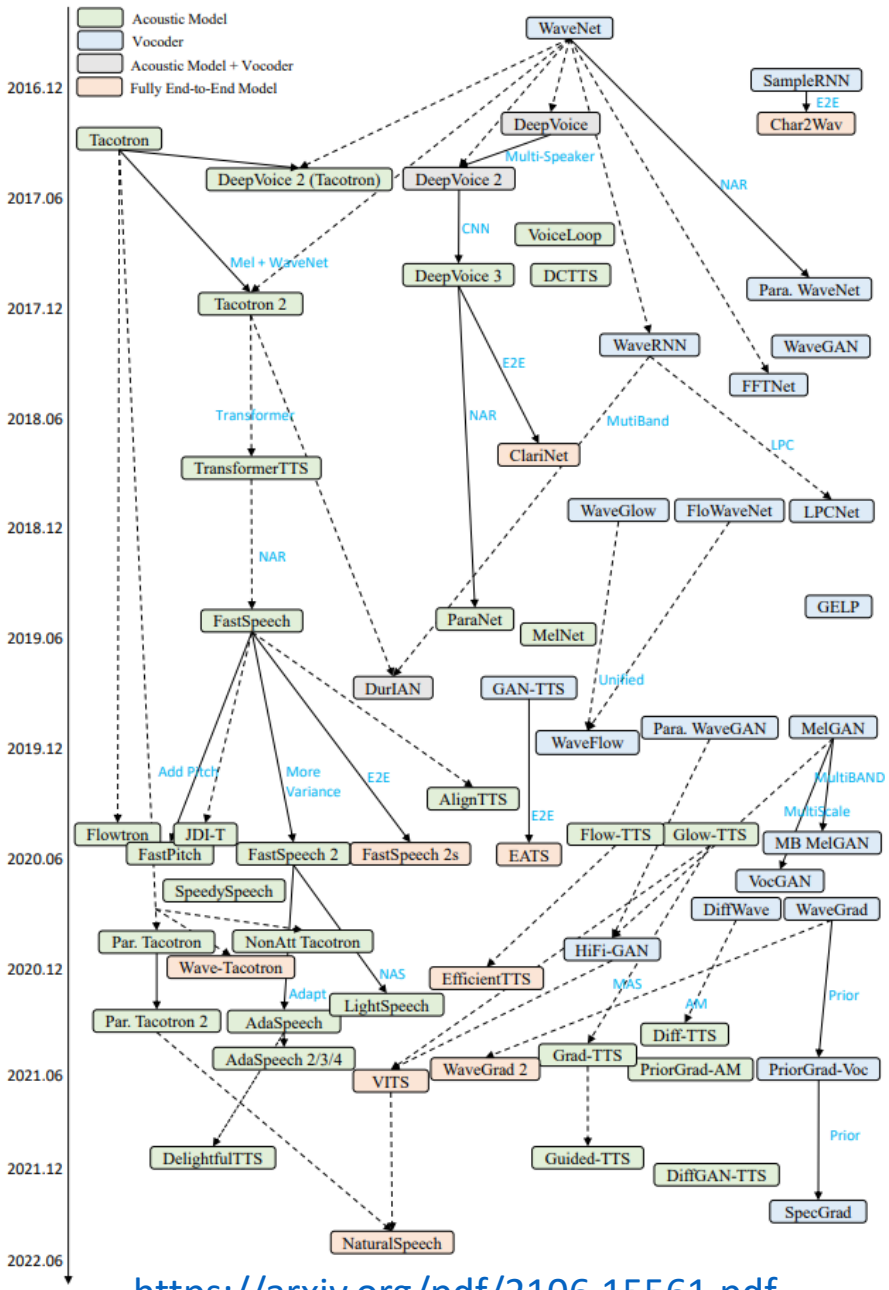
# Typical Neural TTS Pipeline

- Text analysis, acoustic model, and vocoder




- Text analysis: text → linguistic features
- Acoustic model: linguistic features → acoustic features
- Vocoder: acoustic features → speech

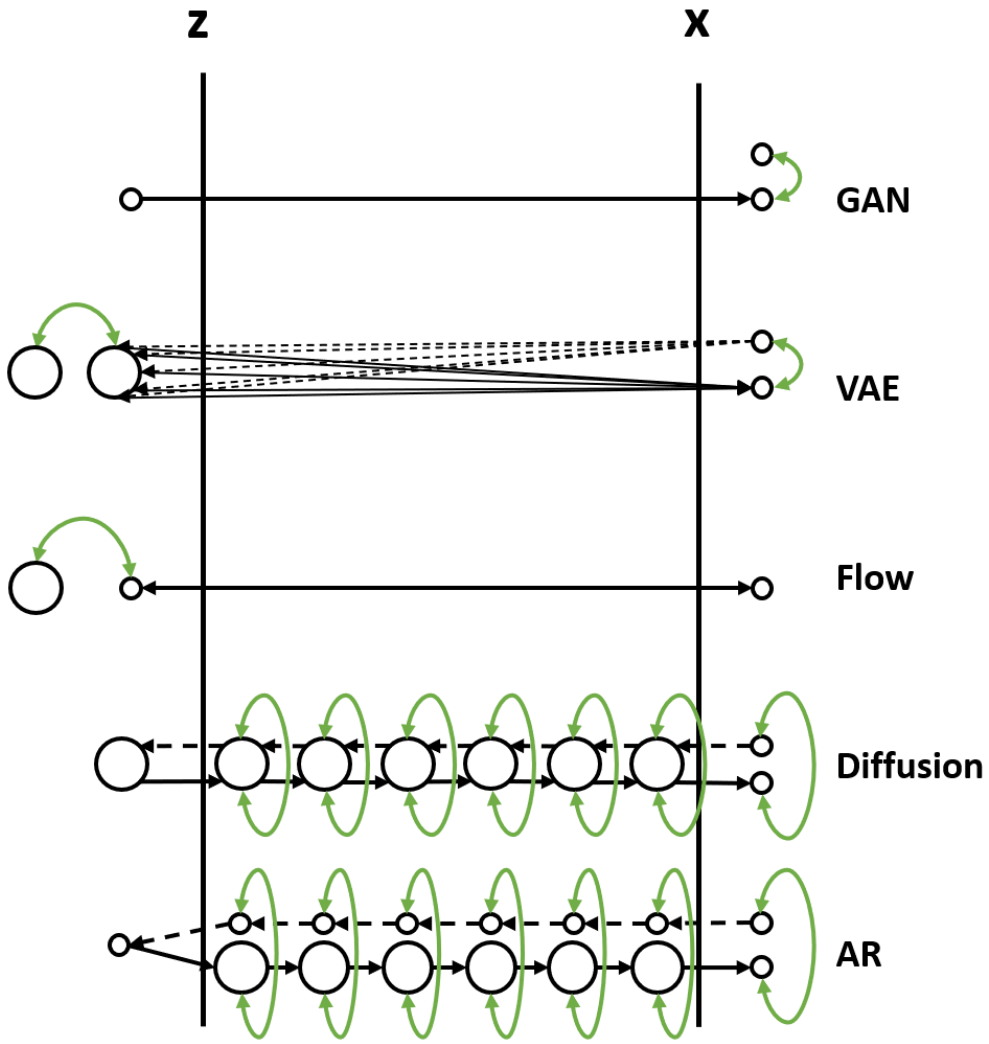
- Linguistic features: phoneme, prosody features
- Acoustic features: **mel-spectrogram, discrete token, latent vector**




https://arxiv.org/pdf/2106.15561.pdf

# AR vs NAR in Neural TTS

- Generative models can be classified in AR/NAR
  - AR: Autoregressive
  - NAR: GAN, VAE, Flow, Diffusion (Flow Matching)
- Difference between AR and NAR (Diffusion)
  - How to factorize data?
    - AR: along time axis
    - NAR: along noise level
  - How to determine alignment/duration?
    - AR: implicitly
    - NAR: explicitly
  - Iteration steps
    - AR: sequence length
    - NAR: flexible




https://zhuanlan.zhihu.com/p/591881660

# The Battle Between AR and NAR

| | Acoustic Model | | Vocoder | |
|---|---|---|---|---|
| Timeline | AR | NAR | AR | NAR |
| 2017.06 | Tacotron | | WaveNet | |
| 2017.12 | Tacotron 2, DeepVoice 3 | | | Par. WaveNet |
| 2018.06 | | | WaveRNN | |
| 2018.12 | Transformer TTS | | LPCNet | WaveGlow |
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| 2020.12 | | | | DiffWave, WaveGrad, HiFiGAN |
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| 2021.12 | | VITS | | **SoundStream** |
| 2022.06 | | NaturalSpeech | | BigVGAN |
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# Trends of the AR/NAR Battle

- Trend 1: NAR dominates Vocoder (Codec)
- Trend 2: NAR shows advantage in acoustic model before the LLM era
- Trend 3: LLMs revive the AR/NAR battle

# Explanation of Trend 1&2

- Target-Target (T-T) vs Target-Source (T-S) dependency
  - T-T: dependency among target tokens
  - T-S: dependency on source tokens
- Difficulty of AR/NAR
  - If T-T > T-S → more information is needed from target tokens → NAR is more difficult
  - Vice versa
- Connection to multi-modality
  - Multi-modality: P(x|y) is not single-modal, not one-one mapping
    - e.g., “Thank You” → “Vielen Dank” or “Danke”
  - If T-S dominates, P(x|y) is more single-modal, a source token will have one definite mapping
  - If T-T dominates, P(x|y) is multi-modal, a source token will have multiple mappings

# T-S Dependency

| Type of T-S Dependency | Task | Alignment |
|---|---|---|
| Target has correspondence with source | Speech Enhancement | **Inherent** alignment |
| | Voice Conversion | |
| | Text to Speech | **Duration/Attention** alignment |
| | Singing Voice Synthesis | **MusicScore** alignment |
| | Speech Recognition | **CTC/Transducer/Attention** alignment |
| Target is a minor change of source | Text Error Correction | Locate the minor changes |
| | Text Style Transfer | Content not changes but style changes |
| Target is a translation of source | Machine Translation | **Attention** alignment |
| Target is implicitly correlated to source | Dialogue Generation | **Semantic** alignment |
| | Image Generation | **Semantic** alignment |

# T-T Dependency

| Type of T-T Dependency | Task | Description |
|---|---|---|
| Text | Machine Translation | Discrete tokens in languages are **contextualized**, explained mutually.<br>**Strong mutual dependency** |
| | Text Summarization | |
| | Text Error Correction | |
| | Text Style Transfer | |
| | Dialogue Generation | |
| | Speech Recognition | |
| Speech and Image | Text to Speech | For continuous signal like speech/sound/image, they depends on the concept, like speech frames depend on a word, image pixel depend on a class.<br>**Weaker mutual dependency** |
| | Singing Voice Synthesis | |
| | Image Generation | |

# T-T/T-S Dependency and NAR Difficulty

| Modality | Task | Source | Target | T-T vs T-S | Difficulty of NAR |
|---|---|---|---|---|---|
| Text Generation | Machine Translation | Source language | Target language | ≈ | ***** |
| | Text Summarization | Long text | Short Summarization | ≈ | ***** |
| | Dialogue Generation | Dialogue | Response | > | ****** |
| | Text Error Correction | Error Text | Correct Text | < | *** |
| | Text Style Transfer | Source Text | Target text | < | *** |
| | Speech Recognition | Speech | Text | ≤ | **** |
| Speech Generation | **Text to Speech** | **Text** | **Speech** | < | *** |
| | Singing Voice Synthesis | Score | Singing Voice | < | ** |
| | Voice Conversion | Source Voice | Target Voice | ≪ | * |
| | Speech Enhancement | Noisy Speech | Clean Speech | ≪ | * |
| Image Generation | Pixel Generation | Class ID | Image Pixel | - | * |
| | Discrete Token Generation | | Image Token | - | ** |

# Explanation of Trend 1&2

- Trend 1: NAR dominates Vocoder (Codec)
- Trend 2: NAR shows advantage in acoustic model before the LLM era

| | TTS (Overall) | Vocoder | Acoustic Model |
|---|---|---|---|
| Target | Signal<br>Not Symbol | Continuous Signal<br>(Perceptual) | Content/Prosody/Timbre/Acoustic<br>(Semantic) |
| T-T Dependency | Weaker Than Text | Short-term, Low-level, Local | Long-term, High-level, Global |
| T-S Dependency | 1-1 Correspondence | Frame-level alignment | Duration/Attention Alignment |
| NAR Difficulty | **Easier than ASR/NMT** | **Very Easy, NAR Dominates** | **Easy, NAR Shows Advantage Before LLM Era** |

# Lessons Learned From Trend 1&2

- **Lesson 1**: To generate low-level perceptual details, **NAR is preferred**. If T-S has strong dependency, NAR is the best choice.
  - Audio (speech/music/sound): vocoder, codec
  - Image: VAE/VQ-VAE/VQ-GAN
  - Image/audio super-resolution/enhancement
- **Lesson 2**: To generate high-level semantic information, **AR is preferred**. If T-S has no strong correspondence, AR is the best choice.
  - LLMs for text generation
  - Non-autoregressive NMT is a great lesson
- **Lesson 3**: To generate mid-level semantic/acoustic information, **NAR has advantages**, if T-S has strong dependency, and speed/robustness are considered
  - NAR-based acoustic model in TTS, speed/robustness are better than AR-based
  - e.g., FastSpeech 2 vs Transformer TTS

Iterative NAR can also do well in modeling T-T dependency!

# Trend: LLMs Revive the AR/NAR Battle

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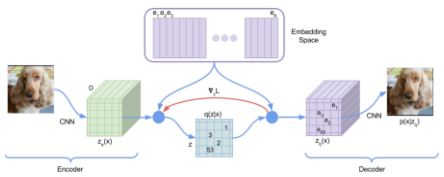
# Lesson 4: The Goal/Paradigm of TTS Has Shifted In the New Era

- Original goal: synthesize intelligible and natural speech
  - Intelligible: **achieved**
  - Natural: quality on limited styles/speakers/languages, **achieved**
- Goal now: natural and human-like
  - Diverse styles/speakers/languages
  - Huge effort to cover so many varieties
    - Prosody/emotion/style: **unlimited** variety
    - Speaker/timbre: **billions** of speakers in the world
    - Content/language: **thousands** of languages
- The paradigm to achieve the new goal
  - **Pre-train** on large-scale/diverse data
  - **Fine-tune** on specific style/speaker/language
  - **Zero-shot/in-context learning** on novel styles/speakers/languages

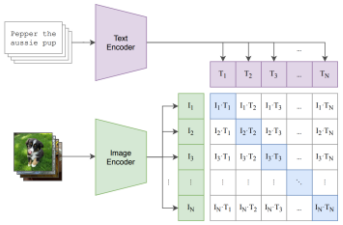
# Recap LLM-based TTS

- Neural data representation/tokenization



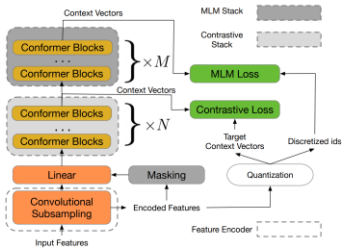

(Image)

**VQ-VAE**



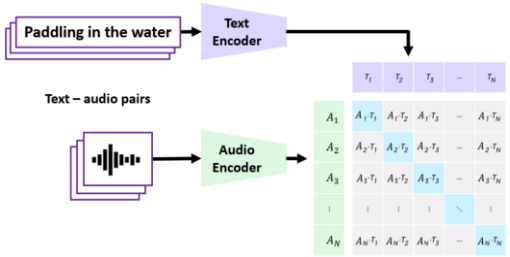

(Text-Image)

**CLIP**



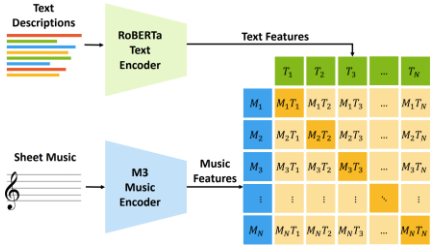

(Audio)

**w2v-BERT**




(Text-Audio)

**CLAP**



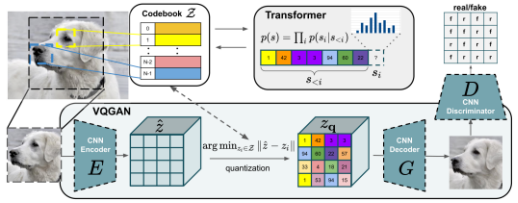

(Text-Audio)

**CLaMP**

2017.11 — 2020.12 — 2021.3 — 2021.7 — 2021.8 — 2021.12 — 2022.6 — 2022.8 — 2023.4

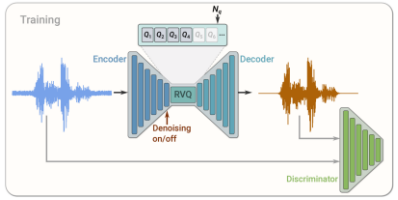
**VQ-GAN**

(Image)



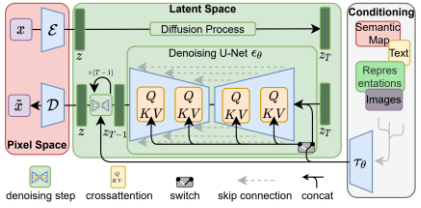

**SoundStream**

(Audio)



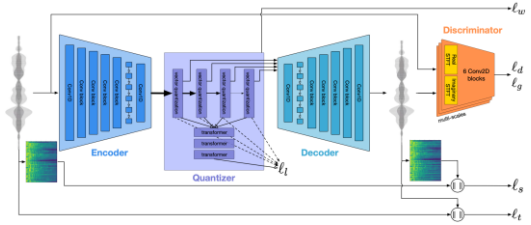

**Stable Diffusion**

(Image)



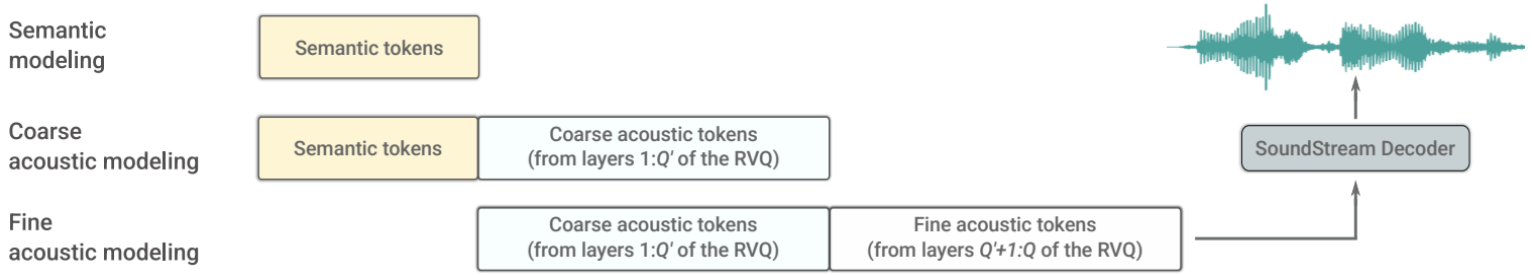

**Encodec**

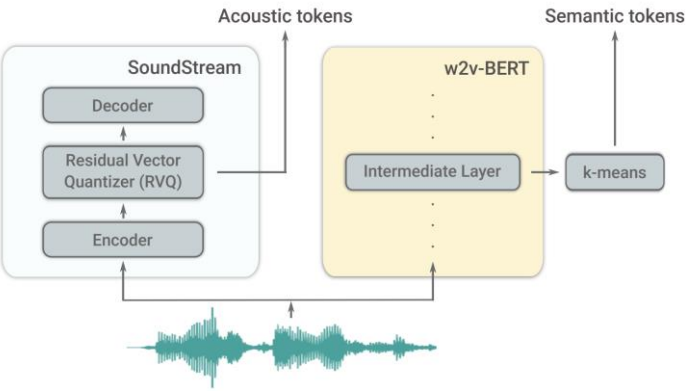
(Audio)

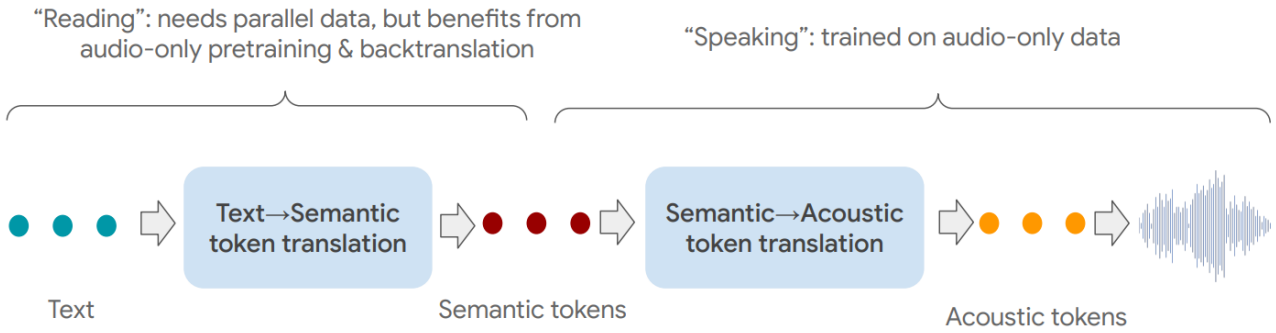


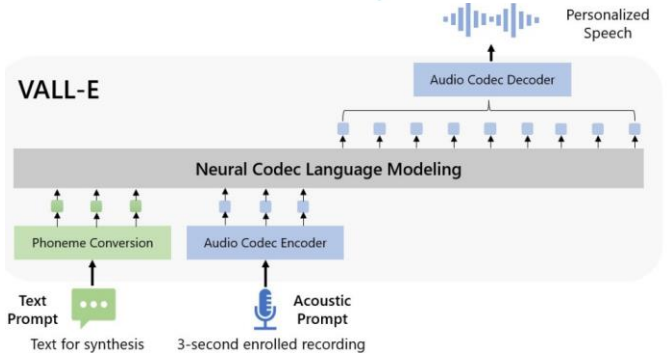
# Recap LLM-based TTS

- Transformer and decoder-only based LLMs
  - **AudioLM**: 1) Semantic, 2) Semantic→Coarse Acoustic, 3) Coarse Acoustic→ Fine Acoustic
  - **SPEAR-TTS**: 1) Text → Semantic Tokens, 2) Semantic → Acoustic
  - **VALL-E**: 1) Text→Acoustic 1st, 2) Acoustic 1st→Acoustic 2nd -8th (NAR)



# Lesson 5: Data/Model Scaling (Out)Weigh Domain Knowledge

- With LLMs and data/model scaling, AR show competitiveness against with NAR
  - Prior domain knowledge (**duration alignment**) show advantages before the LLM era
  - Simple data/modeling scaling (**hundreds of thousands or millions of hours**) weigh or outweigh
  - **Inspirations from other areas (i.e., LLMs) can bring new variables in the battle that was originally going to be lost**
- Perspective
  - Practitioners in TTS: research or product
  - Practitioners in language/speech, audio domain, multimodality

# Lesson 6: The AR/NAR Battle Is Not A Zero-Sum Game

| | AR | NAR |
|---|---|---|
| Models | • AudioLM<br>• VALL-E<br>• SPEAR-TTS<br>• UniAudio | • NaturalSpeech 2<br>• SoundStorm<br>• Mega-TTS<br>• VoiceBox |
| Pros | • Stand on the shoulder of LLMs (e.g., in-context learning, scalability)<br>• Diverse/expressive (sampling) | • Stable/Robust<br>• Fast inference<br>• Control/Disentangle |
| Cons | • Not stable/robust (severe in 0-shot)<br>• Slow inference<br>• Long sequence (complex pipeline) | • Over-smoothness (fidelity, prosody) and less diversity<br>• Complicated alignment process |

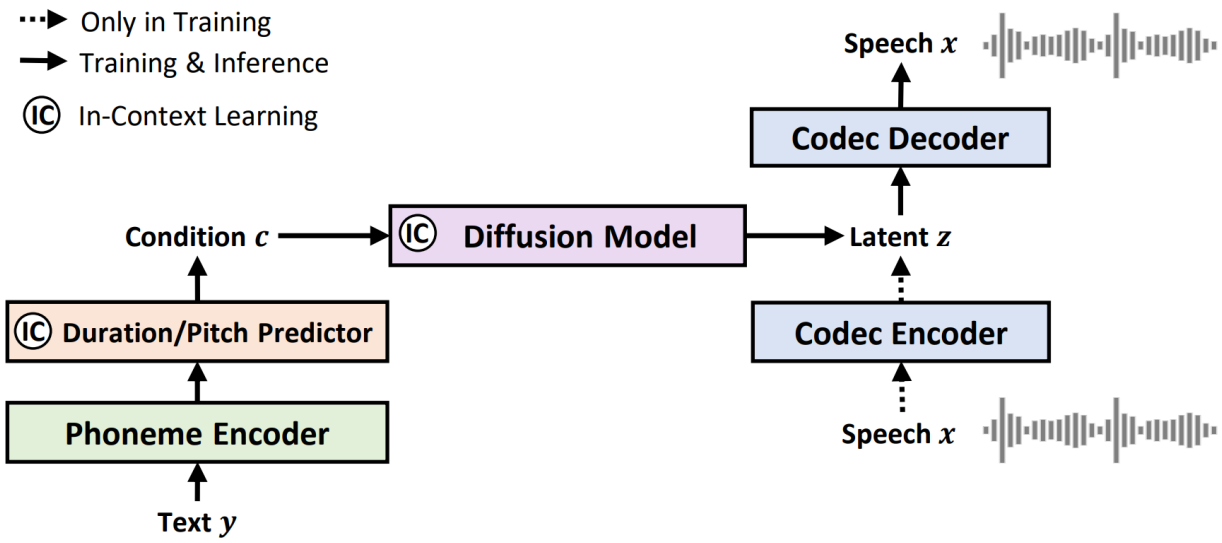
| Difference | AR | NAR | Impact |
|---|---|---|---|
| Data Factorization | Along time axis | Along noise level | |
| Alignment/duration | Implicitly | Explicitly | Stable/Robust, Flexible |
| Iteration steps | Sequence length | Flexible | Fast |

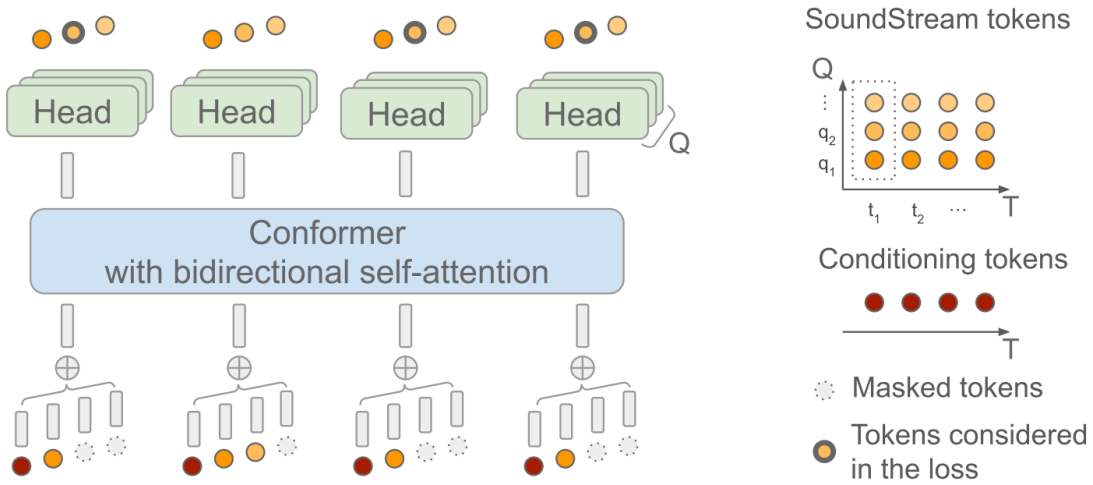
# Lesson 6: The AR/NAR Battle Is Not A Zero-Sum Game

- AR/LLM-based and NAR-based TTS models have different application scenarios
  - AR-based has better **diversity, prosody, expressiveness, and flexibility** than NAR model
  - NAR is better in **speed and robustness**
  - After single-speaker finetuning, **AR models also has few bad cases**, although loses zero-shot capabilities
  - NAR is better in **disentanglement and control** (timbre, prosody, etc)
  - Combine AR and NAR: **semantic-level AR + perceptual-level NAR**

# Lesson 7: Tokenization/Sampling Is Critical for Diversity/Expressiveness

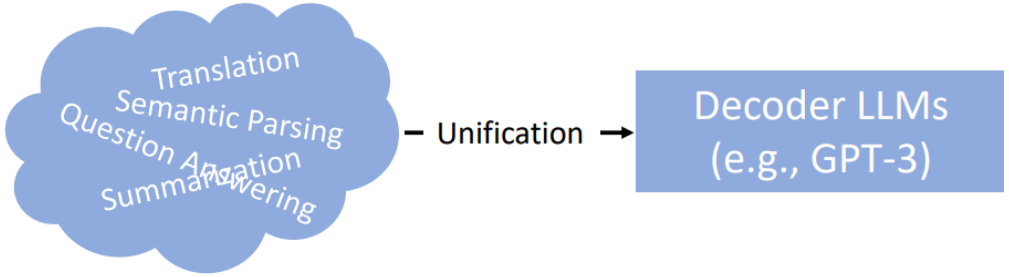
- Tokenization: softmax and cross-entropy
  - Classification to model diverse distribution and support sampling, instead of regression (GAN, VAE, Flow, Diffusion)
  - **Not only benefit for AR but also NAR** (NAR can model discrete tokens)
  - e.g., NaturalSpeech 2 (latent diffusion model with $L_{ce_rvq}$ loss) and SoundStorm



# Lesson 8: Think Outside The Box: The Real Competition May Not Come From Within The Field

- The advantage of LLMs is **scalability and flexibility**, instead of perfect performance on every single task
  - Do not care winning or losing battles but care the war!
- A lesson from NLP




NLP

**Summarization is (Almost) Dead**

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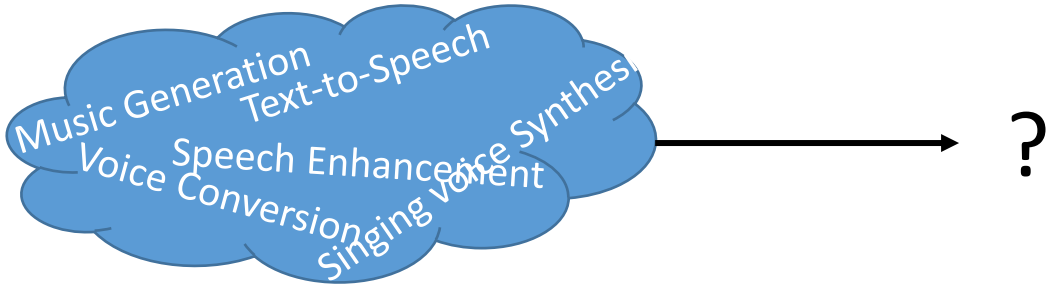
{gaomingqi, wanxiaojun}@pku.edu.cn

- How about speech/audio?



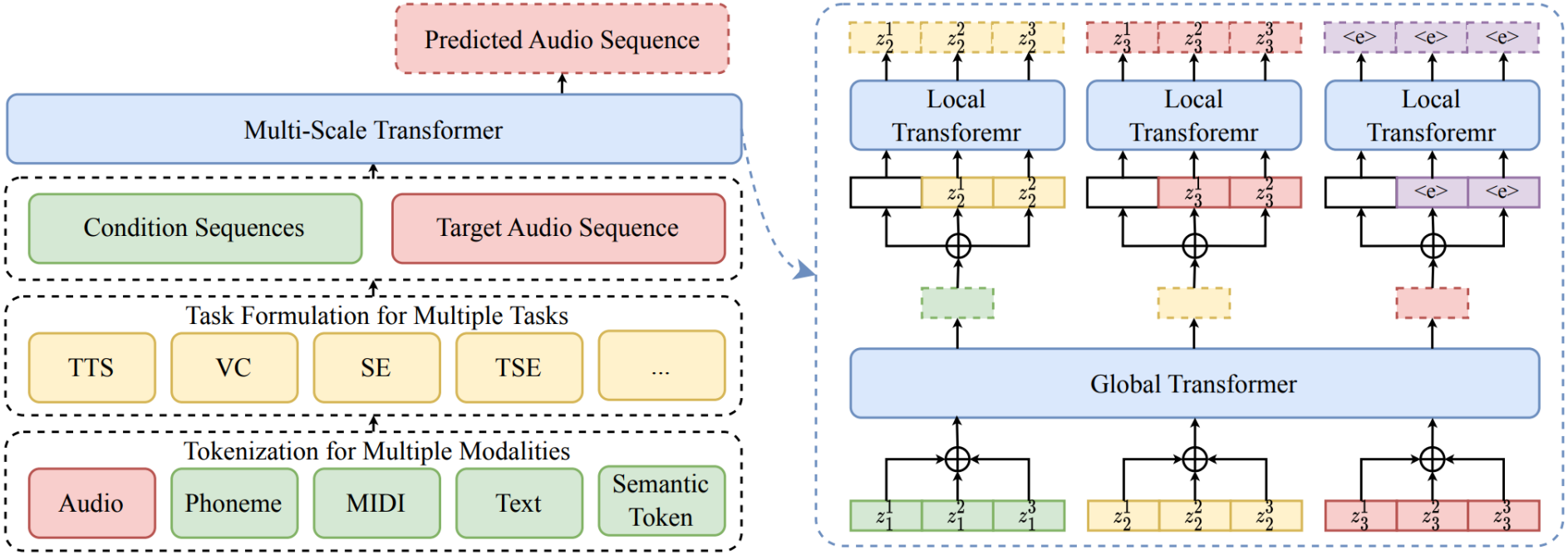
# Lesson 8: Think Outside The Box: The Real Competition May Not Come From Within The Field

- UniAudio: Unify all audio (speech, singing, music, sound) generation tasks in a single LLM
  - Task formulation: concatenate condition-target as a single sequence
  - e.g., <start> <audio_task> <text_start> text_sequence <text_end> <audio_start> audio_sequence <audio_end> <end>

| Task | Conditions | Audio Target |
|---|---|---|
| Text-to-Speech (TTS) (Wang et al., 2023a) | phoneme, speaker prompt | speech |
| Voice Conversion (VC) ♣ (Wang et al., 2023e) | semantic token, speaker prompt | speech |
| Speech Enhancement (SE) ♣ (Wang et al., 2023b) | noisy speech | speech |
| Target Speech Extraction (TSE) ♣ (Wang et al., 2018) | mixed speech, speaker prompt | speech |
| Singing Voice Synthesis (SVS) (Liu et al., 2022) | phoneme (with duration), speaker prompt, MIDI | singing |
| Text-to-Sound (Sound) (Yang et al., 2023c) | textual description | sounds |
| Text-to-Music (Music) (Agostinelli et al., 2023) | textual description | music |
| Audio Edit (A-Edit) ♣◇ (Wang et al., 2023d) | textual description, original sounds | sounds |
| Speech dereverberation (SD) ♣◇ (Wu et al., 2016) | reverberant speech | speech |
| Instruct TTS (I-TTS)◇ (Guo et al., 2023) | phoneme, textual instruction | speech |
| Speech Edit (S-Edit) ◇ (Tae et al., 2021) | phoneme (with duration), original speech | speech |

# Lesson 8: Think Outside The Box: The Real Competition May Not Come From Within The Field

- UniAudio: Unify all audio (speech, singing, music, sound) generation tasks in a single LLM



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| Model | TTS | VC | SE | TSE | SVS | Sound | Music | A-Edit | SD | I-TTS | S-Edit |
|---|---|---|---|---|---|---|---|---|---|---|---|
| YourTTS (Casanova et al., 2022) | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| VALL-E (Wang et al., 2023a) | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| MusicLM (Wang et al., 2023a) | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ |
| SPEARTTS (Kharitonov et al., 2023) | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| NaturalSpeech2 (Shen et al., 2023) | ✓ | ✓ | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Make-A-Voice (Huang et al., 2023b) | ✓ | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Maga-TTS (Jiang et al., 2023) | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| VoiceBox (Le et al., 2023) | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| AudioLDM2 (Liu et al., 2023b) | ✓ | ✗ | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ |
| SpeechX (Wang et al., 2023c) | ✓ | ✗ | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| UniAudio (ours) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

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- The advantage of LLMs is **scalability and flexibility**, instead of perfect performance on every single task
  - Do not care winning or losing battles but care the war!
  - UniAudio-like work will dominate the whole audio tasks, not merely TTS or generation
  - Universal task support (speech/singing/music/sound, understanding/generation), next word prediction, scaling law, in-context learning, prompting

# Lessons Learned

- **Lesson 1**: To generate low-level perceptual detail, NAR is preferred. If T-S has strong dependency, NAR is the best choice
- **Lesson 2**: To generate high-level semantic information, AR is preferred. If T-S has no strong dependency, AR is the best choice
- **Lesson 3**: To generate mid-level semantic/acoustic information, NAR has advantages, if T-S has strong dependency, and speed/robustness are considered
- **Lesson 4**: The goal/paradigm of TTS has shifted in the new era
- **Lesson 5**: Data/model scaling (out)weigh domain knowledge
- **Lesson 6**: The AR/NAR battle is not a zero-sum game
- **Lesson 7**: Tokenization/sampling is critical for diversity/expressiveness
- **Lesson 8**: Think outside the box: the real competition may not come from within the field

# Tips From These Lessons

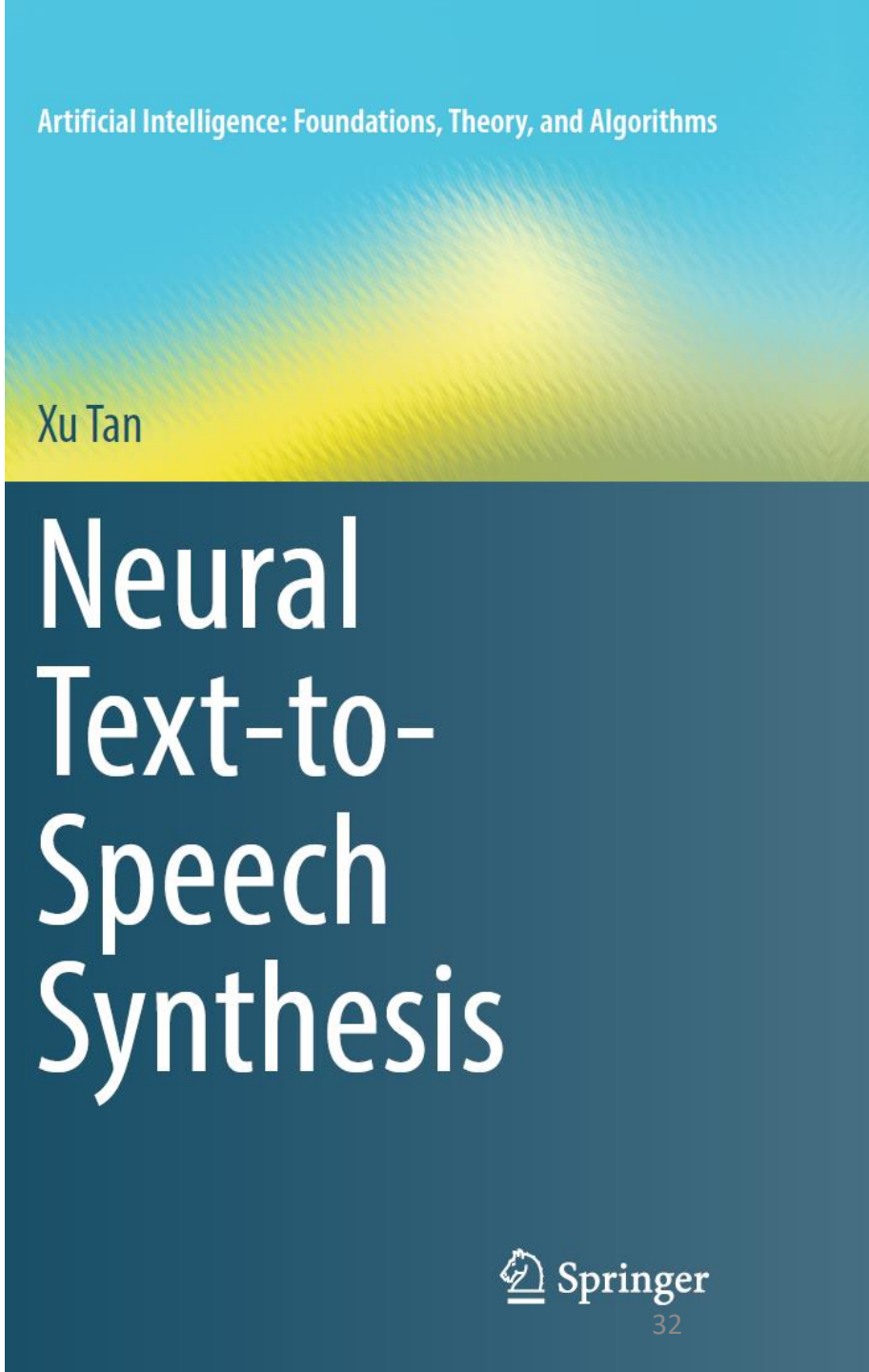
- **Tip 1**: **Choose AR/NAR** according to your scenarios (more signal/perceptual or semantic/contextual, fast inference, streaming, high-quality single speaker, zero-shot, stableness, scalability?)
- **Tip 2**: **Exploit NAR**, e.g., tokenization/sampling, disentanglement/control, stable zero-shot
- **Tip 3**: **Explore AR**, beyond speech synthesis, ChatGPT moment in audio domain
- **Tip 4**: **Scale** data/model/task, explore the unknow

# Thanks

A book on “*Neural Text-to-Speech Synthesis*”

published by Springer!

https://link.springer.com/book/9789819908264



# Thank You!

https://www.microsoft.com/en-us/research/people/xuta/
https://speechresearch.github.io/
tan-xu.github.io